

176. Physical Medium Attachment (PMA) sublayer with RS-FEC symbol multiplexing, type 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R

176.1 Overview

176.1.1 Scope

This clause specifies the Physical Medium Attachment (PMA) sublayer that uses RS-FEC symbol multiplexing for 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R Physical Layer implementations.

When necessary to differentiate the symbol-multiplexing PMA (SM-PMA) types defined in this clause from the bit-multiplexing PMA (BM-PMA) types defined in Clause 120 and Clause 173, the term SM-PMA is used. Within this clause, the term PMA refers specifically to an SM-PMA.

Three types of 200GBASE-R SM-PMA are defined in this clause: 8:1 PMA, 1:8 PMA, and 1:1 PMA.

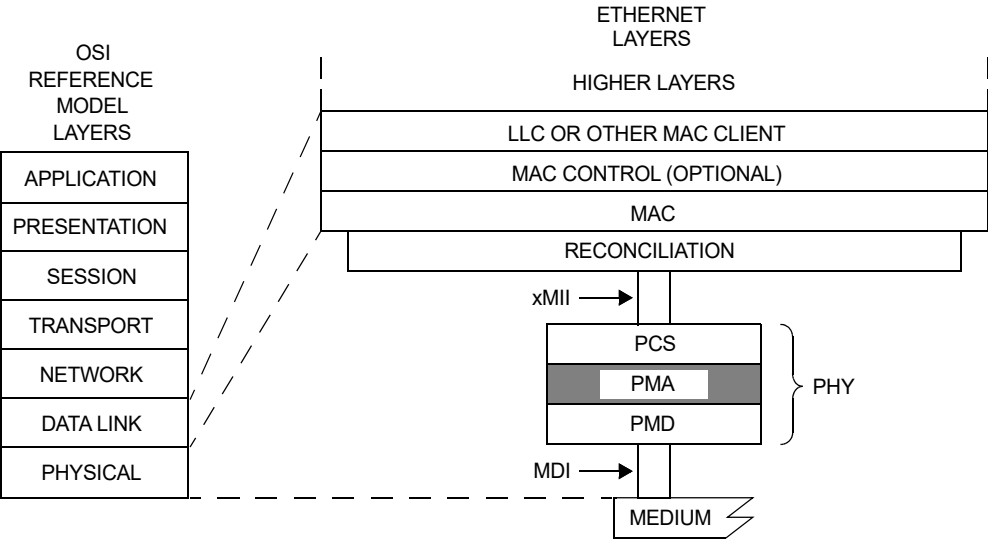
Three types of 400GBASE-R SM-PMA are defined in this clause: 16:2 PMA, 2:16 PMA, and 2:2 PMA.

Three types of 800GBASE-R SM-PMA are defined in this clause: 32:4 PMA, 4:32 PMA, and 4:4 PMA.

Four types of 1.6TBASE-R SM-PMA are defined in this clause: 16:8 PMA, 8:16 PMA, 8:8 PMA, and 16:16 PMA.

176.1.2 Relationship of the PMA to the ISO OSI reference model

Figure 176–1 shows the relationship of the PMA sublayer, along with other Ethernet sublayers, to the ISO Open System Interconnection (OSI) reference model.



xMII = GENERIC MEDIA INDEPENDENT INTERFACE
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 176–1—200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R SM-PMA relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

176.1.3 Conventions

The following terms are defined for use within this clause as follows:

BM-PMA

This term refers to a 200GBASE-R, 400GBASE-R, or 800GBASE-R PMA which provides multiplexing by bits (rather than RS-FEC symbols) of PCS lanes as defined in Clause 120 for 200GBASE-R and 400GBASE-R, and in Clause 173 for 800GBASE-R.

m:n PMA

This term refers collectively to the 200GBASE-R 8:1 PMA, 400GBASE-R 16:2 PMA, 800GBASE-R 32:4 PMA, and 1.6TBASE-R 16:8 PMA.

n:m PMA

This term refers collectively to the 200GBASE-R 1:8 PMA, 400GBASE-R 2:16 PMA, 800GBASE-R 4:32 PMA, and 1.6TBASE-R 8:16 PMA.

n:n PMA

This term refers collectively to the 200GBASE-R 1:1 PMA, 400GBASE-R 2:2 PMA, 800GBASE-R 4:4 PMA, 1.6TBASE-R 8:8 PMA, and 1.6TBASE-R 16:16 PMA.

PMAL

This term refers to a PMA lane operating at 212.5 Gb/s.

SM-PMA

This term refers to a 200GBASE-R, 400GBASE-R, 800GBASE-R, or 1.6TBASE-R PMA which provides multiplexing by RS-FEC symbols (rather than bits) of PCS lanes as defined in this clause.

symbol-pair

A symbol-pair is defined as two adjacent RS-FEC symbols (for example, on a PCS lane) where the first symbol in the pair is from RS-FEC codeword A and the second symbol is from RS-FEC codeword B. Symbol-pairs are used in the 200GBASE-R, 400GBASE-R and 800GBASE-R symbol-multiplexing PMAs.

symbol-quartet

A symbol-quartet is defined as four adjacent RS-FEC symbols (for example, on a PCS lane) where the first symbol in the quartet is from RS-FEC codeword A, the second symbol is from RS-FEC codeword B, the third symbol is from RS-FEC codeword C, and the fourth symbol is from RS-FEC codeword D. Symbol-quartets are used in the 1.6TBASE-R symbol-multiplexing PMAs.

xAUI-n

This term refers collectively to 200GAUI-1, 400GAUI-2, 800GAUI-4, 1.6TAUI-8, and 1.6TAUI-16, where “n” is the number of physical lanes.

xBASE-R

This term refers collectively to 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R.

xMII

This term refers collectively to 200GMII, 400GMII, 800GMII, and 1.6TMII.

xXS

This term refers collectively to 200GXS, 400GXS, 800GXS, and 1.6TXS.

176.1.4 Summary of PMA functions

The following principal functions are provided (when required) by the PMA in both the transmit and receive directions:

- Adapting data from PCS lanes (PCSLs) to one or more PMALs
- Symbol-level multiplexing and demultiplexing
 - Alignment to RS-FEC symbols using the PCSL alignment markers
 - Delay alternating PCSLs by two RS-FEC codewords for 200GBASE-R and 400GBASE-R PMAs
 - Four-way RS-FEC codeword interleaving on all PMALs
- PAM4 encoding and decoding
- Optional local and remote loopbacks
- Test-pattern generation and detection

The m:n PMAs and n:m PMAs primarily provide RS-FEC symbol multiplexing and demultiplexing functions between PCSLs and PMALs. PCSL formats are defined by the xBASE-R PCS sublayer in Clause 119, Clause 172, and Clause 175.

An SM-PMA does not support bit multiplexing interfaces. To convert between bit multiplexing and symbol multiplexing interfaces, a BM-PMA and an SM-PMA are placed back-to-back with each PMA demultiplexing to PCSLs in between. For example, to convert from a 200GAUI-2 interface to a 200GAUI-1 interface, a 200GBASE-R 2:8 BM-PMA (see Clause 120) is placed next to a 200GBASE-R 8:1 SM-PMA.

See Annex 176B for a complete description of xBASE-R PMA instantiations using the SM-PMAs defined in this clause. PMA instantiations for xBASE-R PHYs are described in 176B.5 through 176B.8. The special cases that require back-to-back PMA sublayers are described in 176B.3.

The PMA transmit clock is passed from the interface above the PMA to the interface below in the transmit direction, and the PMA receive clock is passed from the interface below the PMA to the interface above in the receive direction.

176.1.5 PMA sublayer positioning

An implementation may use one or more PMA sublayers to adapt the number and rate of the PCS lanes to the number and rate of the PMD lanes. The number of PMA sublayers required depends on the partitioning of the functionality for a particular implementation. Several examples are illustrated in Annex 176B in Figure 176B–1, Figure 176B–2, Figure 176B–3 and Figure 176B–4.

Table 176–1 through Table 176–4 describe the possible PMA sublayer positioning.

Table 176–1—xBASE-R m:n PMA positioning

Sublayer or interface above PMA	Sublayer or interface below PMA
xBASE-R PCS, xBASE-R BM-PMA, or 1.6TAUI-16 ^a	xAUI-n, xBASE-R Inner FEC, or xBASE-R n-lane PMD
DTE xXS	xAUI-n

^a When x = 1.6T.

Table 176–2—xBASE-R n:m PMA positioning

Sublayer or interface above PMA	Sublayer or interface below PMA
xAUI-n	PHY xXS, xBASE-R BM-PMA, or 1.6TAUI-16 ^a
800GAUI-4	800GBASE-LR1 Inner FEC

^a When x = 1.6T.

Table 176–3—xBASE-R n:n PMA positioning^a

Sublayer or interface above PMA	Sublayer or interface below PMA
xAUI-n	xAUI-n, xBASE-R Inner FEC, or xBASE-R n-lane PMD

^a For all n:n PMAs with the exception of the 1.6TBASE-R 16:16 PMA.

Table 176–4—1.6TBASE-R 16:16 PMA positioning

Sublayer or interface above PMA	Sublayer or interface below PMA
1.6TBASE-R PCS, DTE 1.6TXS, or 1.6TAUI-16	1.6TAUI-16
1.6TAUI-16	PHY 1.6TXS

176.2 PMA service interface

The PMA service interface is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-R and 400GBASE-R, in 169.3 for 800GBASE-R, and in 174.3 for 1.6TBASE-R.

The PMA service interfaces are illustrated in Figure 176–2, Figure 176–12 and Figure 176–13.

The PMA service interface primitives (when applicable) are summarized as follows:

```
PMA:IS_UNITDATA_i.request(tx_symbol)
PMA:IS_SIGNAL.request(SIGNAL_OK)
PMA:IS_UNITDATA_i.indication(rx_symbol)
PMA:IS_SIGNAL.indication(SIGNAL_OK)
```

When the client sublayer is an xBASE-R PCS, DTE xXS or a BM-PMA, each instance of tx_symbol and rx_symbol takes on one of two values: 0 or 1. When there is an xAUI-n above the PMA, each instance of tx_symbol and rx_symbol takes on one of four values: 0, 1, 2, or 3 using PAM4 signaling.

In the transmit direction, the m:n PMAs receive m parallel symbol streams from the client sublayer as PMA:IS_UNITDATA_0:($m-1$).request. In the receive direction, the m:n PMAs send m parallel symbol streams to the client sublayer as PMA:IS_UNITDATA_0:($m-1$).indication. The nominal data rate per stream at the client interface for the 200GBASE-R 8:1, 400GBASE-R 16:2, and 800GBASE-R 32:4 PMAs is 26.5625 Gb/s. The nominal data rate per stream at the client interface for the 1.6TBASE-R 16:8 PMA is 106.25 Gb/s.

In the transmit direction, the n:m PMAs receive n parallel symbol streams from the client sublayer as PMA:IS_UNITDATA_0:($n-1$).request. In the receive direction, the n:m PMAs send n parallel symbol streams to the client sublayer as PMA:IS_UNITDATA_0:($n-1$).indication. The nominal data rate per stream at the client interface for all n:m PMAs is 212.5 Gb/s.

In the transmit direction, the n:n PMAs receive n parallel symbol streams from the client sublayer as PMA:IS_UNITDATA_0:($n-1$).request. In the receive direction, the n:n PMAs send n parallel symbol streams to the client sublayer as PMA:IS_UNITDATA_0:($n-1$).indication. The nominal data rate per stream at the client interface for the 200GBASE-R 1:1, 400GBASE-R 2:2, 800GBASE-R 4:4, and 1.6TBASE-R 8:8 PMAs is 212.5 Gb/s. The nominal data rate per stream at the client interface for the 1.6TBASE-R 16:16 PMA is 106.25 Gb/s.

The PMA receives signal status information from the client sublayer using the PMA:IS_SIGNAL.request primitive. The PMA provides signal status information to the client sublayer using the PMA:IS_SIGNAL.indication primitive. For the m:n and n:m PMAs the SIGNAL_OK parameter at the

client interface is set by the signal indication logic (SIL) according to Table 176–5. For the n:n PMAs, the SIGNAL_OK parameter at the client interface is set to the value of the received SIGNAL_OK parameter from the sublayer below the PMA (*inst:IS_SIGNAL.indication(SIGNAL_OK)*).

Table 176–5—PMA:IS_SIGNAL.indication(SIGNAL_OK) generation

<i>inst:IS_SIGNAL.indication</i> ^a SIGNAL_OK	align_status_mux ^b or all_locked_demux ^c	PMA:IS_SIGNAL.indication SIGNAL_OK
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

^a From the sublayer below the PMA.

^b For n:m PMAs (see 176.4.4.2.1).

^c For m:n PMAs (see 176.4.4.2.1).

176.3 Service interface below PMA

The service interface below the PMA is an instance of the inter-sublayer service interface defined in 116.3 for 200GBASE-R and 400GBASE-R, in 169.3 for 800GBASE-R, and in 174.3 for 1.6TBASE-R.

The service interface primitives (when applicable) are summarized as follows:

inst:IS_UNITDATA_i.request(tx_symbol)
inst:IS_SIGNAL.request(SIGNAL_OK)
inst:IS_UNITDATA_i.indication(rx_symbol)
inst:IS_SIGNAL.indication(SIGNAL_OK)

When the sublayer below the PMA is a PHY xXS, BM-PMA, or Inner FEC, then each instance of tx_symbol and rx_symbol takes on one of two (bit) values: 0 or 1. When there is an xAUI-n or an xBASE-R PMD below the PMA, then each instance of tx_symbol and rx_symbol takes on one of four values: 0, 1, 2, or 3 using PAM4 signaling.

In the transmit direction, the m:n PMAs send *n* parallel symbol streams to the sublayer below the PMA as *inst:IS_UNITDATA_0:(n–1).request*. In the receive direction, the m:n PMAs receive *n* parallel symbol streams from the sublayer below the PMA as *inst:IS_UNITDATA_0:(n–1).indication*. The nominal data rate per stream at the service interface below the m:n PMAs is 212.5 Gb/s.

In the transmit direction, the n:m PMAs send *m* parallel symbol streams to the sublayer below the PMA as *inst:IS_UNITDATA_0:(m–1).request*. In the receive direction, the n:m PMAs receive *m* parallel symbol streams from the sublayer below the PMA as *inst:IS_UNITDATA_0:(m–1).indication*. The nominal data rate per stream at interface below the PMA for the 200GBASE-R 1:8, 400GBASE-R 2:16 and 800GBASE-R 4:32 PMAs is 26.5625 Gb/s. The nominal data rate per stream at the interface below the 1.6TBASE-R 8:16 PMA is 106.25 Gb/s.

In the transmit direction, the $n:n$ PMAs send n parallel symbol streams to the sublayer below the PMA as *inst:IS_UNITDATA_0:(n-1).request*. In the receive direction, the $n:n$ PMAs receive n parallel symbol streams from the sublayer below the PMA as *inst:IS_UNITDATA_0:(n-1).indication*. The nominal data rate per stream at the interface below the PMA for the 200GBASE-R 1:1, 400GBASE-R 2:2, 800GBASE-R 4:4 and 1.6TBASE-R 8:8 PMAs is 212.5 Gb/s. The nominal data rate per stream at the interface below the PMA for the 1.6TBASE-R 16:16 PMA is 106.25 Gb/s.

The PMA receives signal status information from the sublayer below the PMA using the *inst:IS_SIGNAL.indication* primitive. The PMA provides signal status information to the sublayer below the PMA using the *inst:IS_SIGNAL.request* primitive. For $m:n$ and $n:m$ PMAs, the SIGNAL_OK parameter at the interface below the PMA is set by the SIL according to Table 176–6. For the $n:n$ PMAs, the SIGNAL_OK parameter at the interface below the PMA is set to the value of the received SIGNAL_OK parameter from the sublayer above the PMA (PMA:IS_SIGNAL.request(SIGNAL_OK)).

Table 176–6—*inst:IS_SIGNAL.request*(SIGNAL_OK) generation

PMA:IS_SIGNAL.request ^a SIGNAL_OK	align_status_mux ^b or all_locked_demux ^c	<i>inst:IS_SIGNAL.request</i> ^d SIGNAL_OK
OK	true	OK
OK	false	READY
READY	don't care	READY
IN_PROGRESS	don't care	IN_PROGRESS
FAIL	don't care	FAIL

^a From the sublayer above the PMA.

^b For $m:n$ PMAs (see 176.4.4.2.1).

^c For $n:m$ PMAs (see 176.4.4.2.1).

^d To the service interface below the PMA.

176.4 200GBASE-R 8:1, 400GBASE-R 16:2, 800GBASE-R 32:4 and 1.6TBASE-R 16:8 SM-PMAs

This subclause defines the $m:n$ PMAs using the variables m and n to denote the number of input and output lanes of each PMA, where:

- m is 8 and n is 1 for the 200GBASE-R 8:1 PMA
- m is 16 and n is 2 for the 400GBASE-R 16:2 PMA
- m is 32 and n is 4 for the 800GBASE-R 32:4 PMA
- m is 16 and n is 8 for the 1.6TBASE-R 16:8 PMA

Note that m equals the number of PCSs and n equals the number of PMALs for each xBASE-R $m:n$ PMA.

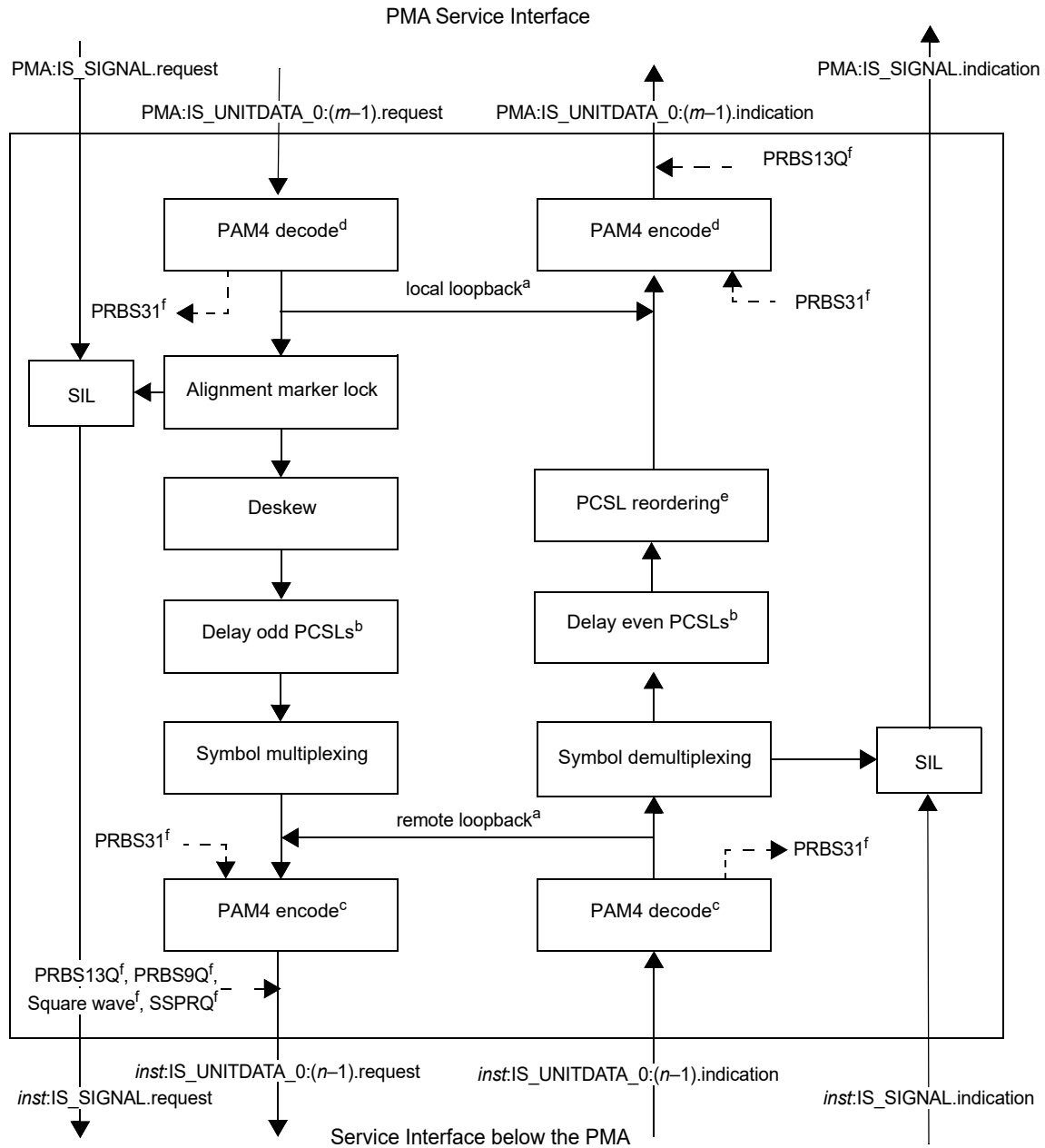
176.4.1 Overview

In the transmit (multiplexing) direction, the $m:n$ PMAs multiplex RS-FEC symbols from m PCSs at the PMA service interface to n PMALs at the service interface below the PMA. In the receive (demultiplexing) direction, the $m:n$ PMAs demultiplex RS-FEC symbols from n PMALs at the service interface below the PMA to m PCSs toward the PMA service interface. The transmit and receive functions are described in

176.4.2 and 176.4.3, respectively. The sublayer or interfaces above and below the m:n PMAs are shown in Table 176–1.

A functional block diagram of the m:n PMAs is shown in Figure 176–2.

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inst: PMA or PMD or FEC
SIL: Signal indication logic

^a Optional

^b If 200GBASE-R, 400GBASE-R or 800GBASE-R

^c If an AUI or PMD is below the PMA

^d If an 1.6TAUI-16 is above the PMA

^e If an 800GBASE-R BM-PMA is above the PMA

^f Required, optional, or not applicable per 176.7.4

$m = 8$ (200GBASE-R), 16 (400GBASE-R), 32 (800GBASE-R), or 16 (1.6TBASE-R)

$n = 1$ (200GBASE-R), 2 (400GBASE-R), 4 (800GBASE-R), or 8 (1.6TBASE-R)

Figure 176–2—200GBASE-R 8:1, 400GBASE-R 16:2, 800GBASE-R 32:4, 1.6TBASE-R 16:8 PMAs functional block diagram

176.4.2 Transmit function

The transmit function of the $m:n$ PMAs processes m input streams representing m PCS lanes to produce n output streams of symbol-multiplexed data running at 212.5 Gb/s.

As shown in Figure 176–2, the transmit function takes m input lanes from the top of the figure and processes the data through the functional blocks on the left-hand side to n output lanes at the bottom of the figure. PAM4 decoding is first provided at the inputs if necessary. The input PCSLs are then identified by searching for the PCS alignment markers utilizing the same alignment marker lock process as used in the corresponding PCS sublayer. This identifies the RS-FEC symbol boundary and the RS-FEC codeword boundary, enabling the PCSLs to then be deskewed and aligned to a multiple-symbol or codeword boundary which, in turn, allows the symbol multiplexing function to produce 212.5Gb/s output lanes with a round-robin pattern of RS-FEC symbols from four different FEC codewords for all MAC data rates supported by this process. At the output, PAM4 encoding is provided if necessary.

The following subclauses describe each of the transmit functional blocks in the transmit (multiplexing) data flow from the PMA service interface to the service interface below the PMA.

176.4.2.1 PAM4 decode

The transmit PAM4 decode is only required if there is a 1.6TAUI-16 above the PMA. The PAM4 encode and decode functions are described in 176.7.1.

176.4.2.2 Alignment marker lock

The alignment marker lock process for each input PCSL operates in the same manner as the xBASE-R PCS alignment marker lock process (see 119.2.5.1, 172.2.5.1, and 175.2.5.1) with the following exception:

- The `restart_lock` variable in Figure 119–12 takes the value of `restart_lock_mux` which is set in the PMA multiplex synchronization state diagram (Figure 176–10).

Once alignment marker lock is achieved on all input lanes and each lane is locked to a unique alignment marker payload sequence, the PCSLs may be deskewed. Synchronization between the PCS lanes is performed by the process specified in the PMA multiplex synchronization state diagram (Figure 176–10).

176.4.2.3 Deskew

The deskew process is performed to align the PCSLs to an appropriate boundary which enables the symbol multiplexing process to interleave RS-FEC symbols from four different FEC codewords in a round-robin pattern on each output lane. The deskew boundary is different for each $m:n$ PMA as described in 176.4.2.3.1, 176.4.2.3.2, and 176.4.2.3.3.

The deskew process is performed in the DESKEW state of the PMA multiplex synchronization state diagram (Figure 176–10). After the deskew process is complete, the RS-FEC symbol boundaries are known and appropriately aligned across all PCSLs.

176.4.2.3.1 4-codeword deskew

The 4-codeword deskew is used for the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs.

The 4-codeword deskew aligns PCSLs to a four codeword boundary relative to the start of the alignment markers across all PCSLs. To achieve 4-codeword deskew, delay is added independently to the PCSL(s) that arrive earlier in time compared to the last arriving PCSL(s) until there is an integer multiple of four RS-FEC codewords between the start of the alignment markers on any two PCSLs.

Since RS-FEC codewords are distributed across PCSLs, after the 4-codeword deskew is complete, the remaining inter-lane skew between the alignment markers of the PCSLs is in multiples of four codewords as shown below, where N is an integer:

- For 200GBASE-R: $N \times 4 \text{ codewords} \times 544 \text{ symbols per codeword} / 8 \text{ PCSLs} = N \times 272 \text{ symbols}$.
- For 400GBASE-R: $N \times 4 \text{ codewords} \times 544 \text{ symbols per codeword} / 16 \text{ PCSLs} = N \times 136 \text{ symbols}$.

NOTE— A specific case of 4-codeword deskew is the removal of all inter-lane skew between PCSL alignment markers. This corresponds to the case of $N = 0$ for all remaining inter-lane skew after deskew is complete.

176.4.2.3.2 20-bit deskew

The 20-bit deskew is used for the 800GBASE-R 32:4 PMA.

The 20-bit deskew aligns PCSLs to a 20-bit boundary (two RS-FEC symbols) relative to the start of the alignment markers across all PCSLs.

To achieve the 20-bit deskew, delay is added independently to the PCSLs that arrive earlier in time compared to the last arriving PCSL(s), until there is an integer multiple of two RS-FEC symbols (20 bits) between the start of the alignment markers of any two PCSLs.

176.4.2.3.3 40-bit deskew

The 40-bit deskew is used for the 1.6TBASE-R 16:8 PMA.

The 40-bit deskew aligns PCSLs to a 40-bit boundary (4 RS-FEC symbols) relative to the start of the alignment markers across all PCSLs. This delay aligns RS-FEC symbol-quartets across the PCS lanes.

To achieve the 40-bit deskew, delay is added independently to the PCSLs that arrive earlier in time compared to the last arriving PCSL(s), until there is an integer multiple of four RS-FEC symbols (40 bits) between the start of the alignment markers of any two PCSLs.

176.4.2.4 Delay odd PCSLs

The odd PCSLs are delayed by a specific number of RS-FEC symbols with respect to the even PCSLs to enable the multiplexing of RS-FEC symbols from four different codewords by the symbol-pair multiplexer.

This delay function is used by the 200GBASE-R 8:1, 400GBASE-R 16:2, and 800GBASE-R 32:4 PMAs, which employ symbol-pair multiplexing, but not by the 1.6TBASE-R 16:8 PMA, which employs symbol-quartet multiplexing.

Odd PCSLs are delayed by a single RS-FEC symbol as described in 176.4.2.4.1. In addition to the single RS-FEC symbol delay, the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs further delay the odd PCSLs by two codewords as described in 176.4.2.4.2.

This results in the following total delay for each odd PCSL:

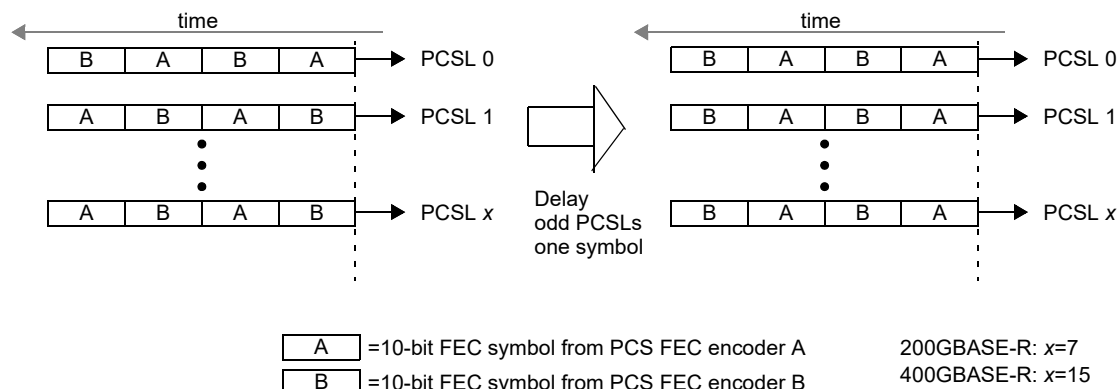
- The 200GBASE-R 8:1 PMA delays each odd PCSL by 137 FEC symbols.
- The 400GBASE-R 16:2 PMA delays each odd PCSL by 69 FEC symbols.
- The 800GBASE-R 32:4 PMA delays each odd PCSL by one FEC symbol.
- The 1.6TBASE-R 16:8 PMA adds no delay to the odd PCS lanes.

176.4.2.4.1 Delay odd PCSLs by one symbol

This delay is performed by the 200GBASE-R 8:1, 400GBASE-R 16:2 and 800GBASE-R 32:4 PMAs.

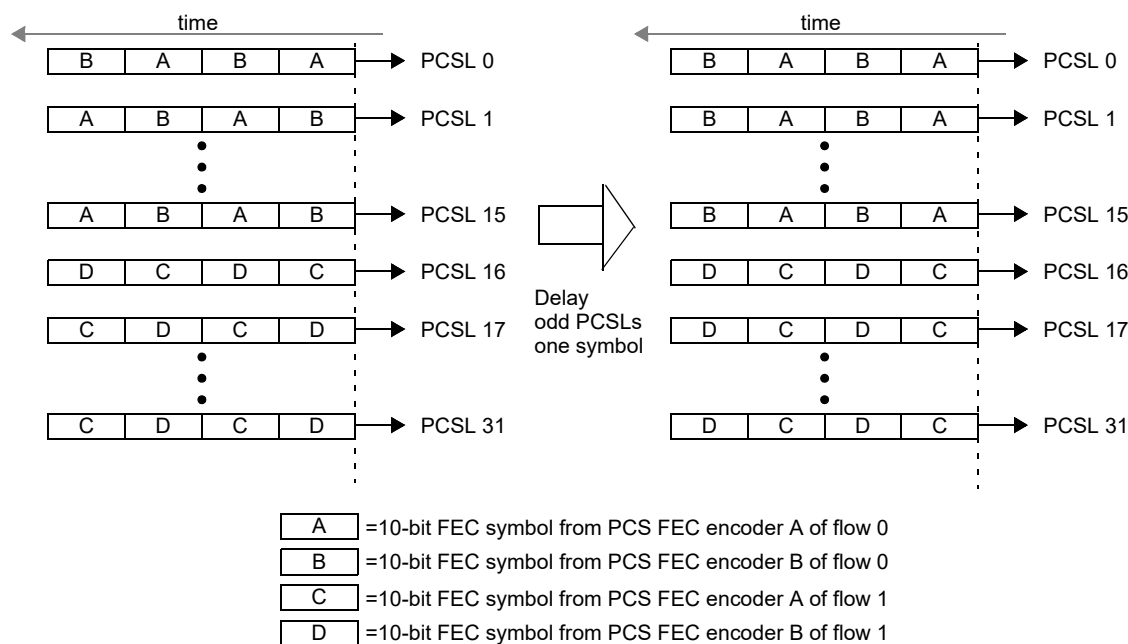
These PMAs delay the odd PCSLs by one RS-FEC symbol (10 bits) to remove the checkerboard pattern created by the distribution of symbols to PCS lanes (see 119.2.4.7). This delay aligns RS-FEC symbol pairs across all PCS lanes. Aligning RS-FEC symbol-pairs eventually enables multiplexing on the symbol-pair boundaries to result in a round-robin pattern among symbols from four different codewords.

Figure 176–3 shows the result of the one symbol delay applied to the odd PCSLs for the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs.



**Figure 176–3—Delay odd PCSLs by one RS-FEC symbol
(200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs)**

Figure 176–4 shows the result of the one symbol delay applied to the odd PCSLs for the 800GBASE-R 32:4 PMA.



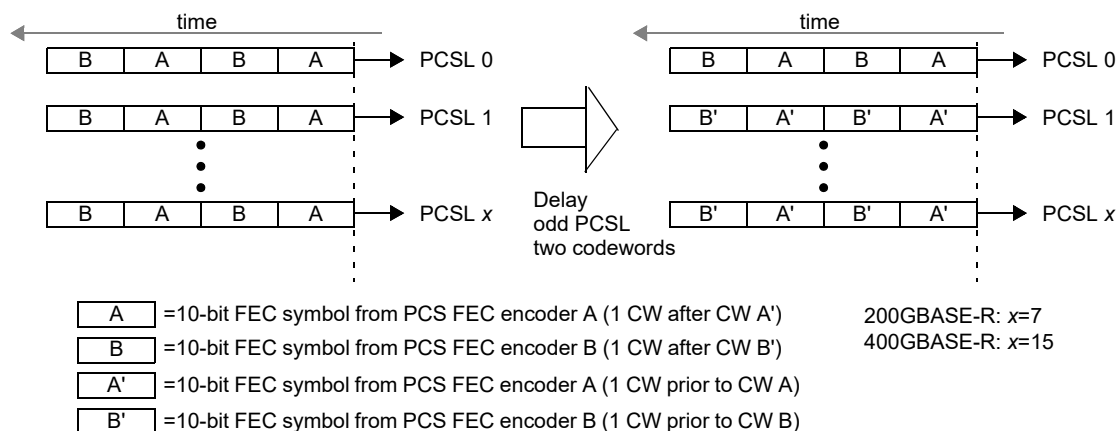
**Figure 176–4—Delay Odd PCSLs by one RS-FEC symbol
(800GBASE-R 32:4 PMA)**

176.4.2.4.2 Delay odd PCSLs by two codewords

This delay is performed for the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs.

For the 200GBASE-R 8:1 PMA, the odd PCSLs are further delayed by 136 RS-FEC symbols (1360 bits), and for the 400GBASE-R 16:2 PMA, the odd PCSLs are further delayed by 68 RS-FEC symbols (680 bits) which is equivalent to adding a delay of two RS-FEC codewords to the odd PCSLs (2 codewords $\times$ 544 symbols per codeword / number of PCS lanes).

Figure 176–5 shows the result of the two codeword delay applied to the odd numbered PCSLs of the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs.



**Figure 176–5—Delay odd PCSLs by two codewords
(200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs)**

176.4.2.5 Symbol multiplexing

The symbol multiplexing function multiplexes symbol-pairs or symbol-quartets. Symbol-pairs are multiplexed in the case of the 200GBASE-R, 400GBASE-R and 800GBASE-R PMAs. Symbol-quartets are multiplexed in the case of the 1.6TBASE-R PMAs. The symbol-pair and symbol-quartet multiplexing functions are defined below.

176.4.2.5.1 Symbol-pair multiplexing

A symbol-pair multiplexer is defined as having eight input lanes and one output lane. The symbol-pair multiplexer shall transmit symbol-pairs from the eight realigned input lanes (PCSLs) in a round-robin manner to the output lane.

For the 200GBASE-R 8:1 PMA and 400GBASE-R 16:2 PMA, the symbol-pair multiplexer input lanes shall consist of four even numbered and four odd numbered PCSLs, and the round-robin multiplexing process shall alternate between even and odd PCS lanes.

For the 800GBASE-R 32:4 PMA, the symbol-pair multiplexer input lanes shall consist of four PCSLs numbered between 0 to 15 (low PCSLs) and four PCSLs numbered between 16 to 31 (high PCSLs). The round-robin multiplexing process shall alternate between low PCSLs and high PCSLs.

Figure 176–6 illustrates an example of an 8:1 symbol-pair multiplexer function for the 200GBASE-R 8:1 and 400GBASE-R 16:2 PMAs starting with deskewed and delayed PCSLs as shown in Figure 176–5. Two

such symbol-pair multiplexing processes are required for the 400GBASE-R 16:2 PMA to service all 16 input lanes. Different output lane orderings are allowed which conform to the above rules for PCSL input lane grouping and ordering.

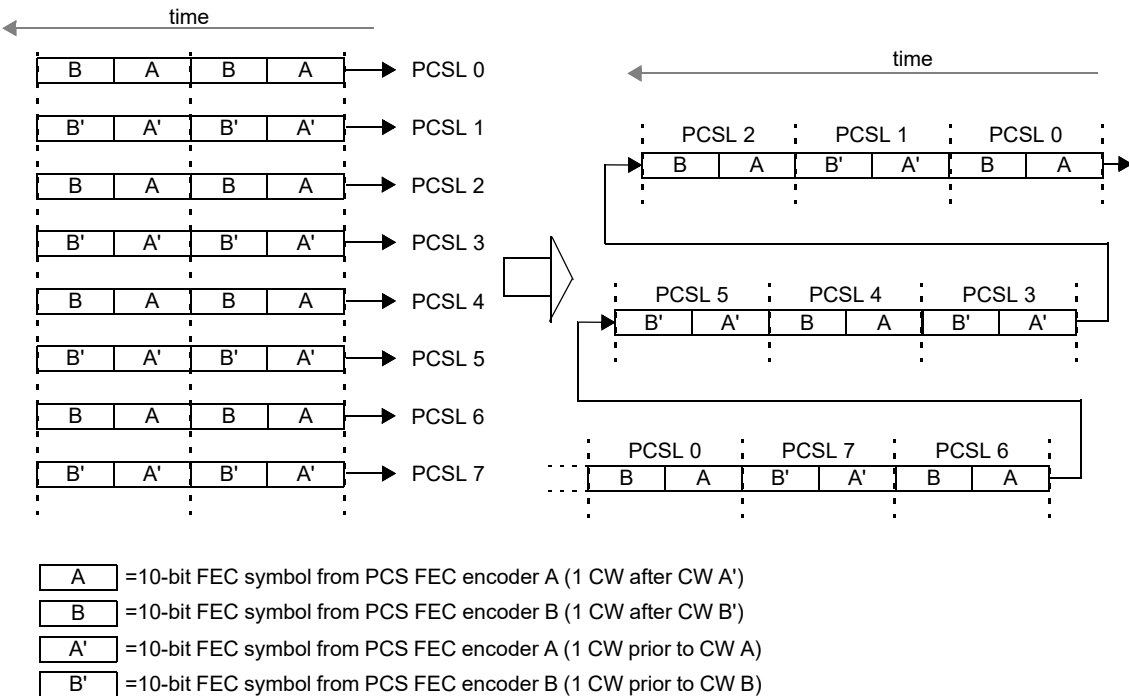


Figure 176-6—Example output lane of the 8:1 symbol-pair multiplexer used in a 200GBASE-R 8:1 PMA or 400GBASE-R 16:2 PMA

Figure 176-7 illustrates an example of an 8:1 symbol-pair multiplexing function for the 800GBASE-R 32:4 PMA starting with deskewed and delayed PCSLs as shown in Figure 176-4. Four similar symbol-pair multiplexing processes are required by 800GBASE-R 32:4 PMA to service all 32 input lanes.

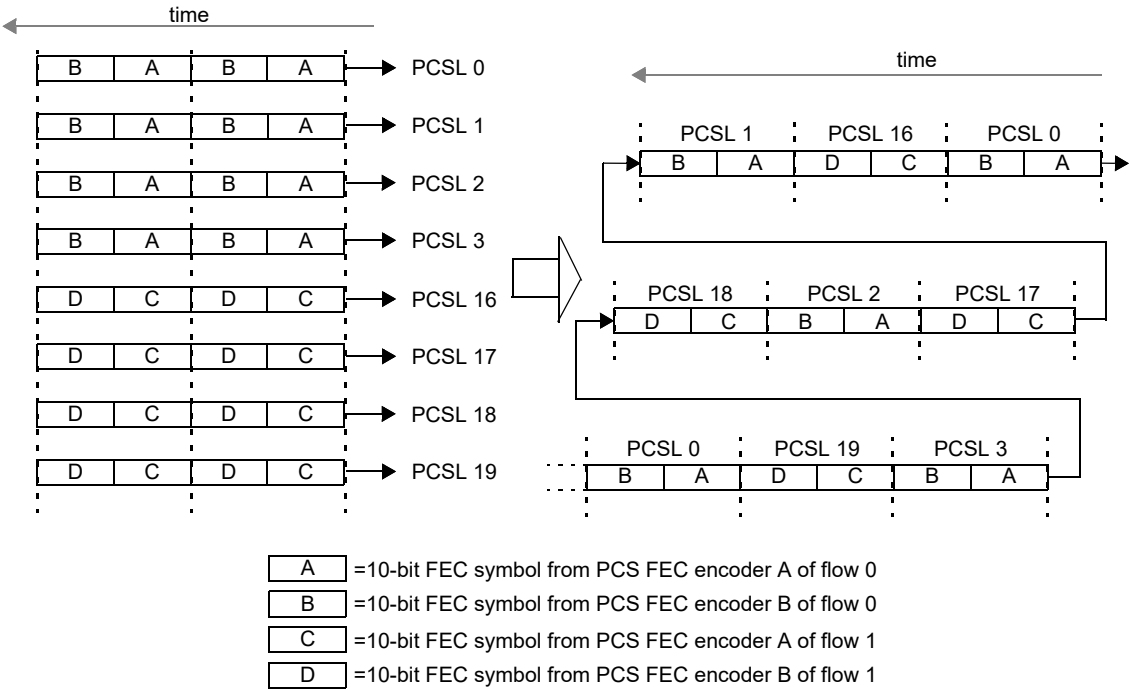


Figure 176-7—Example output lane of a 8:1 symbol-pair multiplexer used in an 800GBASE-R 32:4 PMA

176.4.2.5.2 Symbol-quartet multiplexing

A symbol-quartet multiplexer is defined as having two input lanes and one output lane. The symbol-quartet multiplexer shall transmit symbol-quartets by alternating them from the two realigned input lanes (PCSLs) to the output lane. A symbol-quartet multiplexer may use any two PCSLs as input lanes.

The 1.6TBASE-R 16:8 PMA uses eight symbol-quartet multiplexing processes to multiplex 16 input lanes to eight output lanes.

An example of the symbol-quartet multiplexing process is illustrated in Figure 176-8 starting with two input PCS lanes deskewed to a 40-bit boundary. Note that all 16 PCS lanes come with the same round-robin pattern of four RS-FEC symbols from four separate codewords.

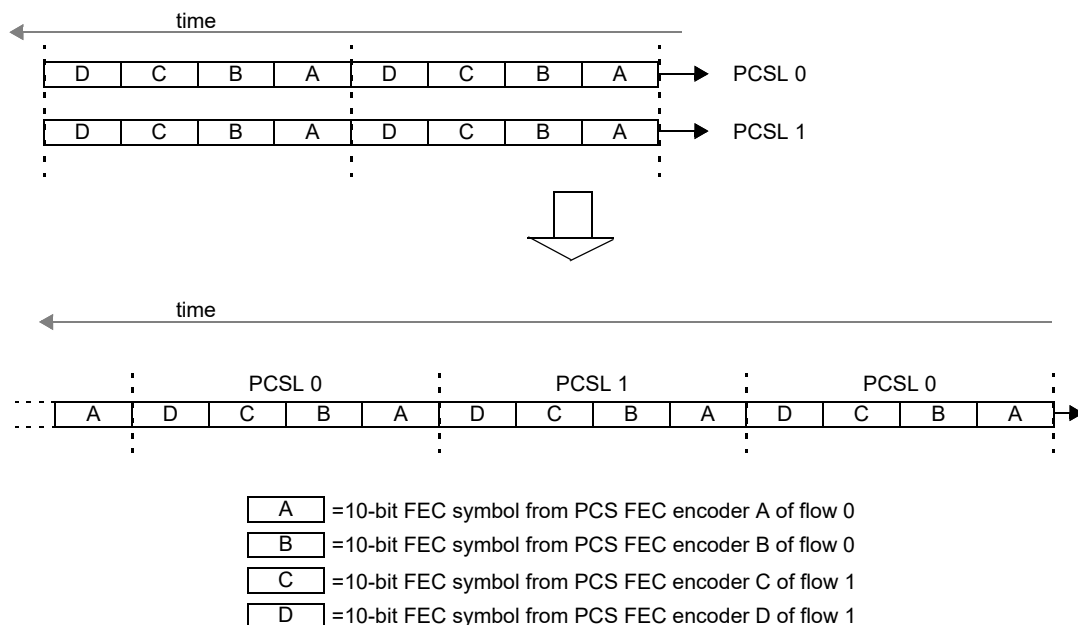


Figure 176-8—Example output lane of a symbol-quartet multiplexer used in a 1.6TBASE-R PMA

176.4.2.6 PAM4 encode

The transmit PAM4 encode process is only required if there is an AUI or PMD below the m:n PMA. The PAM4 encode function is defined in 176.7.1.

176.4.3 Receive function

The PMA receive function of the m:n PMAs processes n input streams of symbol-multiplexed data running at a 212.5 Gb/s to produce m output streams representing m PCS lanes.

As shown in Figure 176-2, the receive function takes n input lanes from the bottom of the figure and processes the data through the functional blocks on the right-hand side to m PCSL output lanes at the top of the figure. PAM4 decoding is provided at the input interface if necessary. The input streams are then demultiplexed at an arbitrary symbol-pair (or symbol-quartet) boundary to produce a set of 8 (or 2) candidate PCSLs from each input stream. These candidate PCSLs are searched for PCS alignment markers using an alignment marker lock process that observes only the symbol boundary given by the symbol demultiplexing function. If all 8 (or 2) PCS lanes successfully locate an alignment marker, then the symbol-pair (or symbol-quartet) boundary is confirmed using the `pmal_locked_demux<y>` indication back to the symbol demultiplexing process, where y is the input PMAL number. When all input PMALs are successfully demultiplexed, the full set of PCS lanes proceeds through the remaining receive functions. The even PCSLs are delayed to match the delay added to the odd PCSLs in the transmit flow, and are then reordered, if required, before being output on the PMA service interface. At the output, PAM4 encoding is provided if necessary.

The following subclauses describe each of the receive functional blocks in the receive (demultiplexing) data flow from the service interface below PMA service interface to the PMA service interface.

176.4.3.1 PAM4 decode

The receive PAM4 decode process is only required if there is an AUI or PMD below the PMA. The PAM4 encode and decode functions are defined in 176.7.1.

176.4.3.2 Symbol demultiplexing

There are two types of symbol demultiplexing, a symbol-pair demultiplexing function and a symbol-quartet demultiplexing function. For the 200GBASE-R, 400GBASE-R and 800GBASE-R PMAs, a 1:8 symbol-pair demultiplex is performed on each input lane to create eight separate PCSs. For the 1.6TBASE-R PMA, a 1:2 symbol-quartet demultiplex is performed on each input lane to create two separate PCSs from each input lane.

The symbol demultiplexing function locates the correct symbol demultiplex boundary and achieves symbol lock on each input PMAL. After all input PMALs from the service interface below the PMA have achieved symbol lock, the PMA then sends a full set of reconstructed PCSs to the next process in the receive function.

176.4.3.2.1 Symbol-pair demultiplexing

Symbol-pair demultiplexing is performed for the 200GBASE-R 8:1, 400GBASE-R 16:2 and 800GBASE-R 32:4 PMAs.

A functional block diagram of a 1:8 symbol-pair demultiplexer, which is composed of a 20-bit demultiplexing function and an alignment marker lock function (see 176.4.3.2.3), is shown in Figure 176–9.

The 20-bit demultiplexer distributes 20 consecutive bits at a time from an input PMAL to each of eight PCS output lanes using a round-robin process. Alignment marker lock is only achieved on all eight PCS lanes if the 20-bit boundary chosen by the 20-bit demultiplexer aligns to the correct symbol-pair boundary. If alignment marker lock is not achieved on all eight PCS lanes within a time interval set by the symbol-pair demultiplexing process, the 20-bit demultiplexer shifts the 20-bit boundary by one bit (a one bit slip), and restarts the alignment lock process on all eight PCS output lanes. This process of a one-bit slip followed by alignment marker search continues until all eight PCS lanes have alignment marker lock using the 20-bit symbol-pair boundary selected by the 20-bit demultiplexer. When all eight PCS output lanes have alignment marker lock, the input PMAL has achieved symbol-pair lock and `pmal_locked_demux<y>` is set to true, where `y` is the input PMAL number.

The PMA shall implement one PMA demultiplex symbol lock process for each input PMAL as specified by the state diagram in Figure 176–11 to acquire symbol-pair lock and establish the symbol-pair boundary for each input PMAL.

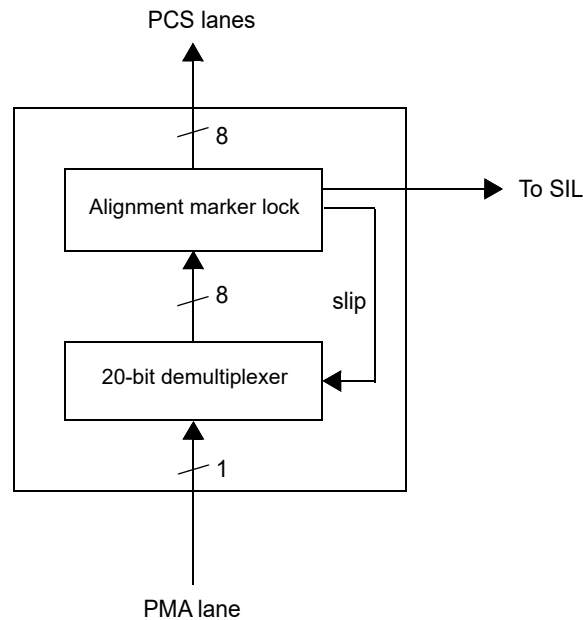


Figure 176–9—1:8 symbol-pair demultiplexer functional block diagram

176.4.3.2.2 Symbol-quartet demultiplexing

Symbol-quartet demultiplexing is performed in the 1.6TBASE-R 16:8 PMA.

Each 1:2 symbol-quartet demultiplexing process is similar to the process defined in 176.4.3.2.1 for the symbol-pair demultiplexing with the exception that it operates on 40-bit boundaries (four RS-FEC symbols) and distributes 40 consecutive bits at a time from an input lane to each of two PCS output lanes.

The 1.6TBASE-R PMA shall implement one PMA demultiplex symbol lock process for each input lane as specified by the PMA demultiplex symbol lock state diagram (Figure 176–11) to establish the symbol-quartet boundary for each input lane.

176.4.3.2.3 Alignment marker lock

The alignment marker lock process for each PCSL after demultiplexing operates in the same manner as the xBASE-R PCS alignment marker lock process (see 119.2.5.1, 172.2.5.1, and 175.2.5.1) with the following exceptions:

- The SLIP function defined in 119.2.6.3.3 slips by 10 bits at a time for the 200GBASE-R 8:1, 400GBASE-R 16:2 and 800GBASE-R 32:4 PMAs. It slips by 40 bits at a time for the 1.6TBASE-R 16:8 PMA.
- The restart_lock variable in Figure 119–12 takes the value of restart_lock_demux<y> which is defined in 176.4.4.2.1 and set by the PMA demultiplex symbol lock state diagram (Figure 176–11).

176.4.3.3 Delay even PCS lanes

Delay is added to the even PCS lanes to match the delay added to the odd PCS lanes in the transmit function described in 176.4.2.4. This removes the artificial skew created by adding delay in the transmit (multiplexing) function.

- The 200GBASE-R 8:1 PMA adds delay equal to 137 FEC symbols to each even PCS lane.
- The 400GBASE-R 16:2 PMA adds delay equal to 69 FEC symbols to each even PCS lane.
- The 800GBASE-R 32:4 PMA adds delay equal to one FEC symbol to each even PCS lane.
- The 1.6TBASE-R 16:8 PMA adds no delay to the even PCS lanes.

176.4.3.4 PCS lane reordering

The PCS lane reordering function is required in the 800GBASE-R PMA when adjacent to an 800GBASE-R BM-PMA which requires its input PCS lanes to be identified by position on the PMA service interface.

PCS lane numbers are identified by the unique portion of the alignment markers, UM0 to UM5 (see 172.2.4.6). PCS lane reordering is performed such that PCS lane x is transmitted as PMA:IS_UNITDATA_x.indication of the PMA service interface, for all $x = 0$ to 31.

176.4.3.5 PAM4 encode

The receive PAM4 encode process is only required if there is a 1.6TAUI-16 above the PMA. The PAM4 encoding function is described in 176.7.1.

176.4.4 State diagrams

This subclause is composed of state diagrams, including the associated definitions of variables, functions, and counters.

176.4.4.1 State diagram conventions

The notation used in the state diagrams follows the conventions of 21.5.

Should there be a discrepancy between a state diagram and descriptive text, the state diagram prevails.

176.4.4.2 State diagram definitions

The 200GBASE-R 8:1 PMA, 400GBASE-R 16:2 PMA, 800GBASE-R 32:4 PMA, and 1.6TBASE-R 16:8 PMA use the alignment marker lock state diagram from Clause 119 (Figure 119–12).

Any constants, variables, functions, or counters used in the alignment marker lock state diagram (Figure 119–12), that are not defined in the following subclauses, shall use the values and definitions for each PMA from the following subclauses:

- For 200GBASE-R 8:1 PMA, use 119.2.6.2
- For 400GBASE-R 16:2 PMA, use 119.2.6.2
- For 800GBASE-R 32:4 PMA, use 172.2.6.2
- For 1.6TBASE-R 16:8 PMA, use 175.2.6.2

176.4.4.2.1 Variables

The following variables are common to all PMA state machines:

PMA_reset

A Boolean variable that is true when set by a management entity and is false otherwise.

reset

A Boolean variable that controls the resetting of the PMA sublayer. It is true whenever a reset is necessary including when initiated by PMA_reset and during power on, and is false otherwise.

The following variables are used in the PMA multiplex synchronization state diagram (Figure 176–10) and the alignment marker lock state diagram (Figure 119–12) when used for the PMA multiplexing function (in the multiplexing direction).

- `align_status_mux`
Boolean variable that indicates the alignment marker lock and deskew processes are complete. Its value is set by the PMA multiplex synchronization state diagram (Figure 176–10).
- `all_locked_mux`
Boolean variable that takes the value of the variable `all_locked` as defined in 119.2.6.2.2.
- `deskew_done_mux`
Boolean variable that is set to true when deskew is complete. Otherwise, it is set to false.
- `deskew_enable_mux`
Boolean variable that is used to start the deskew process. Its value is set by the PMA multiplex synchronization state diagram (Figure 176–10).
- `pcs_lanes_identified_mux`
Boolean variable that is set to true if each input PCS lane is locked to a unique alignment marker sequence identified using the alignment markers in Table 119–1 for 200GBASE-R, Table 119–2 for 400GBASE-R, Table 172–2 and Table 172–3 for 800GBASE-R, or Table 175–4 for 1.6TBASE-R PMAs. It is set to false upon entering the `LOSS_OF_ALIGNMENT` state in the PMA multiplex synchronization state diagram (Figure 176–10).
- `restart_lock_mux`
Boolean variable that indicates the lane synchronization process has failed and is used to restart the alignment marker lock process on all input PCS lanes (see 176.4.2.2). Its value is set by the PMA multiplex synchronization state diagram (Figure 176–10).
- `signal_ok_mux`
Boolean variable that is set based on the most recently received value of the `SIGNAL_OK` primitive on the service interface in the multiplexing direction. It is true if the value was OK and false otherwise.

The following variables are used in the PMA demultiplex symbol lock state diagram (Figure 176–11) and the alignment marker lock state diagram (Figure 119–12) when used for the PMA demultiplexing function (in the demultiplexing direction). The index variable n represents the number of input PMAs.

- `all_locked_demux`
Boolean variable that is set to true when `pmal_locked_demux<y>` is true for all y , where $y = 0$ to $(n-1)$, which indicates all PCS lanes within all PMA lanes have achieved alignment marker lock. Otherwise, this variable is set to false.
- `pcs_lanes_identified_demux`
Boolean variable that is set to true if all demultiplexed PCS lanes are locked to a unique alignment marker sequence identified using the alignment markers in Table 119–1 for 200GBASE-R, Table 119–2 for 400GBASE-R, Table 172–2 and Table 172–3 for 800GBASE-R, or Table 175–4 for 1.6TBASE-R PMAs. It is set to false upon entering the `LOSS_OF_SYMBOL_LOCK` state in the PMA demultiplex symbol lock state diagram (Figure 176–11).
- `pmal_locked_demux<y>`
Boolean variable that is set to true when `amps_lock<x>` is true, as defined in 119.2.6.2.2, for all PCSs within the single input PMA lane y in the demultiplexing direction, and is set to false otherwise. For $y = 0$ to $(n-1)$.
- `restart_lock_demux<y>`
Boolean variable that is used to restart the alignment marker lock processes for the PCSs within

the single input lane y in the demultiplexing direction, where $y = 0$ to $(n-1)$. Its value is set by the PMA demultiplex symbol lock state diagram (Figure 176–11).

signal_ok_demux

Boolean variable that is set based on the most recently received value of the SIGNAL_OK primitive of the service interface in the demultiplexing direction. It is true if the value was OK and false otherwise.

symbol_lock_counter_demux_done< y >

Boolean variable that is set to true when the counter symbol_lock_counter_demux< y > has reached its terminal count, and is set to false when starting the counter (Figure 176–11). For $y = 0$ to $(n-1)$.

176.4.4.2.2 Functions

BIT_SLIP_DEMUX

This function shifts the symbol-pair or symbol-quartet boundary used by the symbol demultiplexing function by one bit. This causes the next candidate symbol-pair or symbol-quartet boundary to be tested by its associated alignment marker lock processes.

SLIP

The SLIP function used in the alignment marker lock process for the PMA multiplexing direction is the same as defined in 119.2.6.3.3.

The SLIP function used in the alignment marker lock process for the PMA demultiplexing direction is defined differently. It slips by ten bits at a time for the 200GBASE-R, 400GBASE-R, and 800GBASE-R PMAs. It slips by 40 bits at a time for the 1.6TBASE-R PMAs. These are the only valid starting locations for alignment markers at the symbol boundary set by the PMA demultiplex symbol lock process.

176.4.4.2.3 Counters

symbol_lock_counter_demux< y >

Counts three alignment marker intervals. The alignment marker interval is defined as 4096 FEC codewords for 200GBASE-R, as 8192 FEC codewords for 400GBASE-R, as 16 384 FEC codewords for 800GBASE-R, and as 32 768 FEC codewords for 1.6TBASE-R. There is one counter associated with each input lane, $y = 0$ to $(n - 1)$.

176.4.4.3 State diagram figures

The PMA multiplexing function (in the multiplexing direction) and the PMA demultiplexing function (in the demultiplexing direction), shall each implement m alignment marker lock processes as depicted in Figure 119–12 where m is the number of input or output lanes (m equals the number of PCSs).

The PMA multiplexing function shall implement a single instance of the PMA multiplex synchronization process as depicted in Figure 176–10. The synchronization process ensures that all PMA input lanes in the multiplexing direction have alignment marker lock, are locked to different alignment markers, and that deskew of all input lanes has been completed to the appropriate boundary for a specific PMA. When these conditions are met, the align_status_mux variable is set to true.

The PMA demultiplexing function shall implement n instances of the PMA demultiplex symbol lock process where n equals the number of input lanes in the demultiplexing direction. Each PMA demultiplex symbol lock process searches for the exact symbol boundary within one input data stream in order to correctly perform the symbol demultiplexing function. Each PMA demultiplex symbol lock process monitors the alignment marker lock processes for the PCSs demultiplexed from within its input lane. When all PCS lanes within its input PMAL have achieved alignment marker lock, the

pmal_locked_demux $\langle y \rangle$ variable is set to true, where y is the input PMAL number. When pmal_locked_demux $\langle y \rangle$ is true for all y , where $y = 0$ to $(n-1)$, the all_locked_demux variable is set to true. When the all_locked_demux and pcs_lanes_identified_demux variables are both true, then the PMA demultiplexing function is able to present correct data on all m output lanes of the service interface. At any time, if all_locked_demux is true and pcs_lanes_identified_demux is false, then restart_lock_demux $\langle y \rangle$ is set to true for all y in order to restart the symbol lock process for all input lanes.

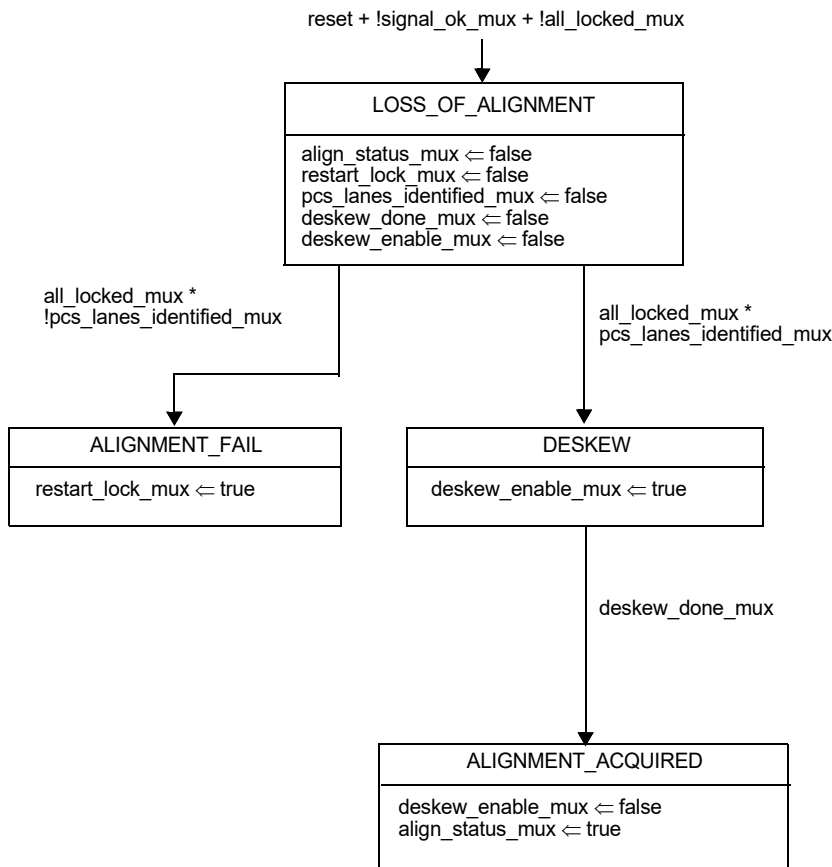


Figure 176–10—PMA multiplex synchronization state diagram

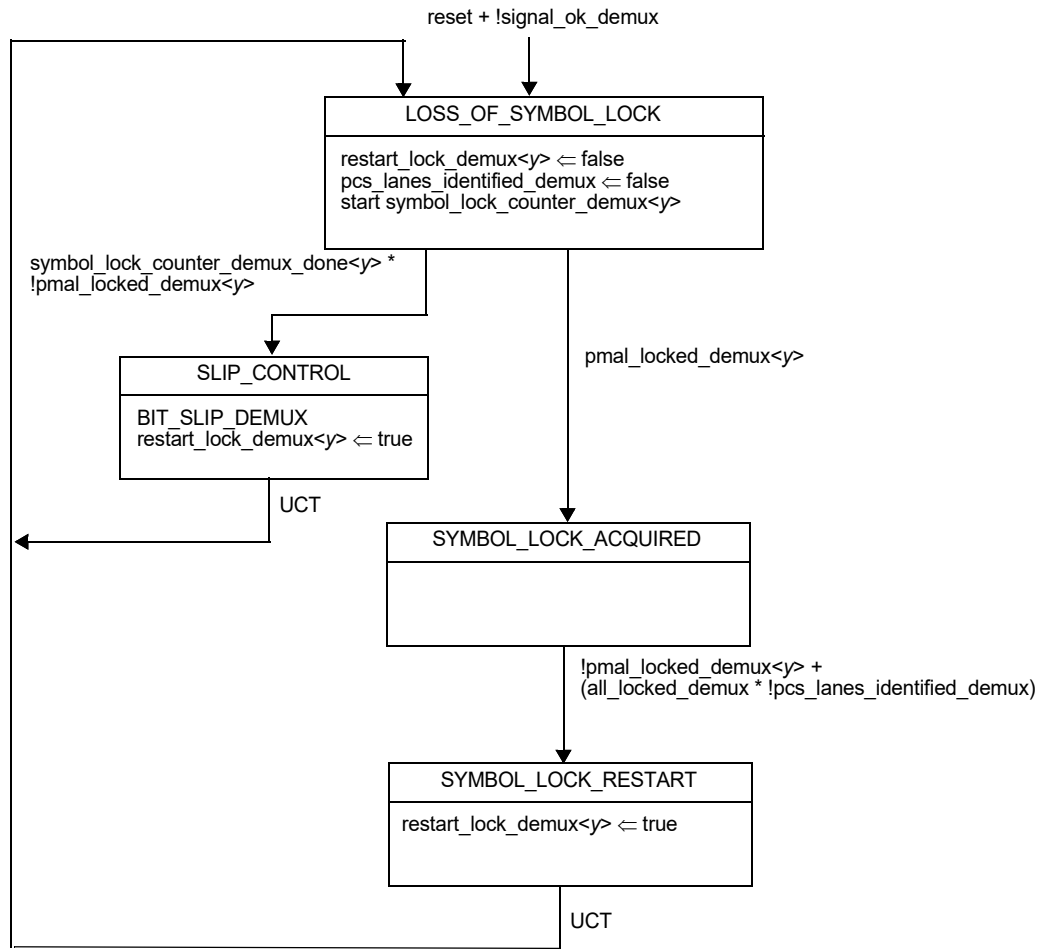


Figure 176–11—PMA demultiplex symbol lock state diagram

176.5 200GBASE-R 1:8, 400GBASE-R 2:16, 800GBASE-R 4:32, 1.6TBASE-R 8:16 SM-PMAs

This subclause defines the $n:m$ PMAs using the variables n and m to denote the number of input and output lanes of the PMA, where:

- n is 1 and m is 8 for the 200GBASE-R 1:8 PMA
- n is 2 and m is 16 for the 400GBASE-R 2:16 PMA
- n is 4 and m is 32 for the 800GBASE-R 4:32 PMA
- n is 8 and m is 16 for the 1.6TBASE-R 8:16 PMA

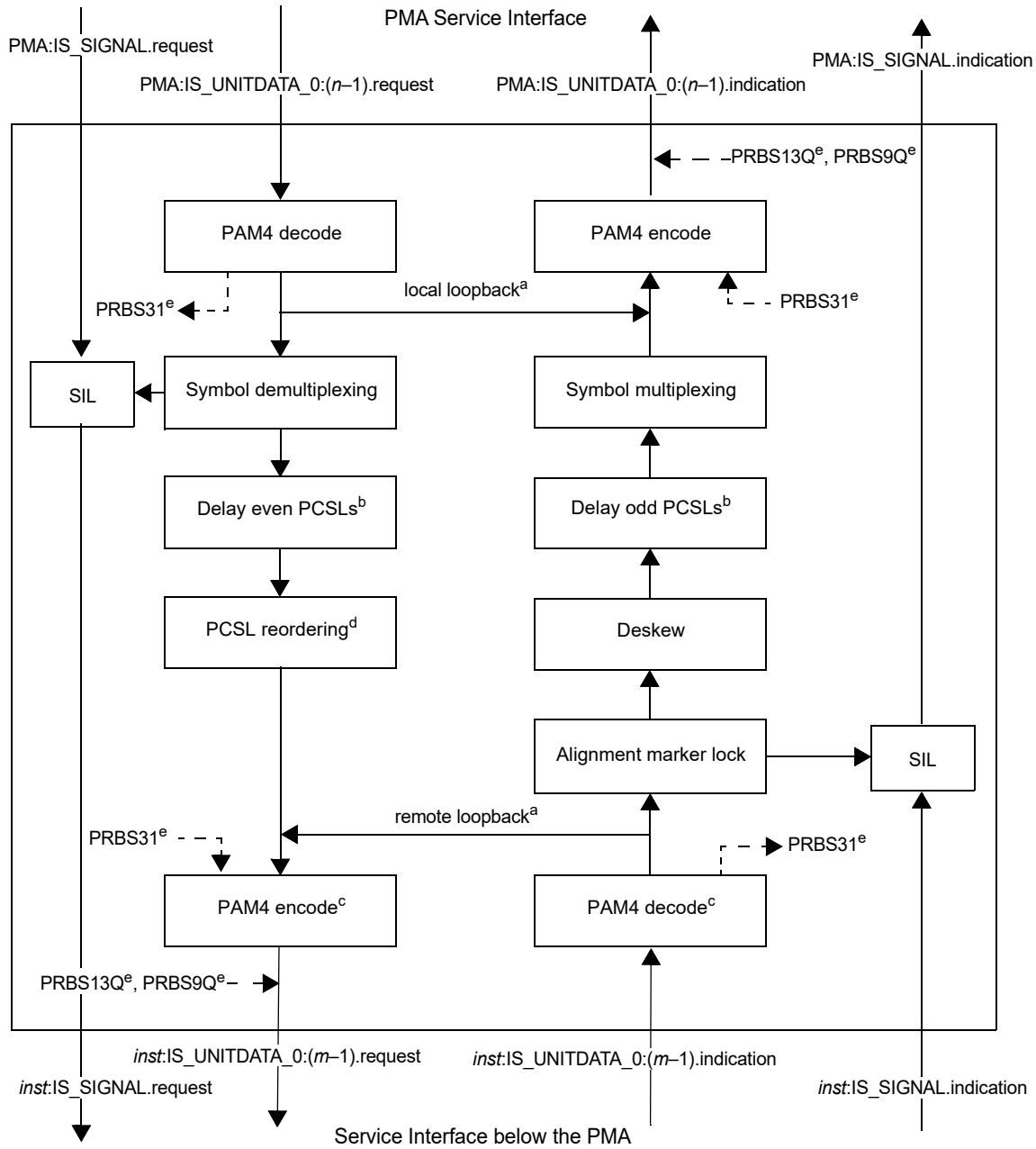
Note that m equals the number of PCSs and n equals the number of PMALs for each xBASE-R $n:m$ PMA.

176.5.1 Overview

In the transmit (demultiplexing) direction, the n : m PMAs demultiplex RS-FEC symbols from n PMALs at the PMA service interface to m PCSs at the service interface below the PMA. In the receive (multiplexing) direction, the n : m PMAs multiplex RS-FEC symbols from m PCSs at the service interface below the PMA to n PMALs at the PMA service interface. The transmit and receive functions are described in 176.5.2 and 176.5.3 respectively. The sublayer or interfaces above and below the n : m PMAs are shown in Table 176–2.

Figure 176–12 provides the functional block diagram of the n : m PMAs.

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inst: PMA or PHY XS or FEC
SIL: Signal indication logic

^a Optional

^b If 200GBASE-R, 400GBASE-R or 800GBASE-R

^c If an 1.6TAUI-16 is below the PMA

^d If an 800GBASE-R BM-PMA is below the PMA

^e Required, optional, or not applicable per 176.7.4

$n = 1$ (200GBASE-R), 2 (400GBASE-R), 4 (800GBASE-R), or 8 (1.6TBASE-R)

$m = 8$ (200GBASE-R), 16 (400GBASE-R), 32 (800GBASE-R), or 16 (1.6TBASE-R)

Figure 176–12—200GBASE-R 1:8, 400GBASE-R 2:16, 800GBASE-R 4:32, 1.6TBASE-R 8:16 PMAs functional block diagram

176.5.2 Transmit function

The transmit function of the $n:m$ PMAs is identical to the receive function of the $m:n$ PMAs (see 176.4.3).

176.5.3 Receive function

The receive function of the $n:m$ PMAs is identical to the transmit function of the $m:n$ PMA (see 176.4.2).

176.6 200GBASE-R 1:1, 400GBASE-R 2:2, 800GBASE-R 4:4, 1.6TBASE-R 8:8, 1.6TBASE-R 16:16 PMAs

This subclause defines the $n:n$ PMAs using the variable n to denote the number of input and output lanes of the PMA, where:

- n is 1 for the 200GBASE-R 1:1 PMA
- n is 2 for the 400GBASE-R 2:2 PMA
- n is 4 for the 800GBASE-R 4:4 PMA
- n is 8 for the 1.6TBASE-R 8:8 PMA
- n is 16 for the 1.6TBASE-R 16:16 PMA

176.6.1 Overview

In the transmit and receive directions, data received on each input lane is mapped to an output lane.

If the interface above or below the PMA is an AUI, PAM4 encoding and decoding is provided by the PMA at that interface.

If the interface above the PMA is an AUI and the interface below is an AUI or a PMD, the Gray mapped PAM4 symbol sequence on each output lane is identical to the Gray mapped PAM4 symbol sequence received on the corresponding input lane, in both transmit and receive directions.

The sublayer or interfaces above and below the $n:n$ PMAs are shown in Table 176–3 and Table 176–4.

Figure 176–13 provides the functional block diagram of the $n:n$ PMAs.

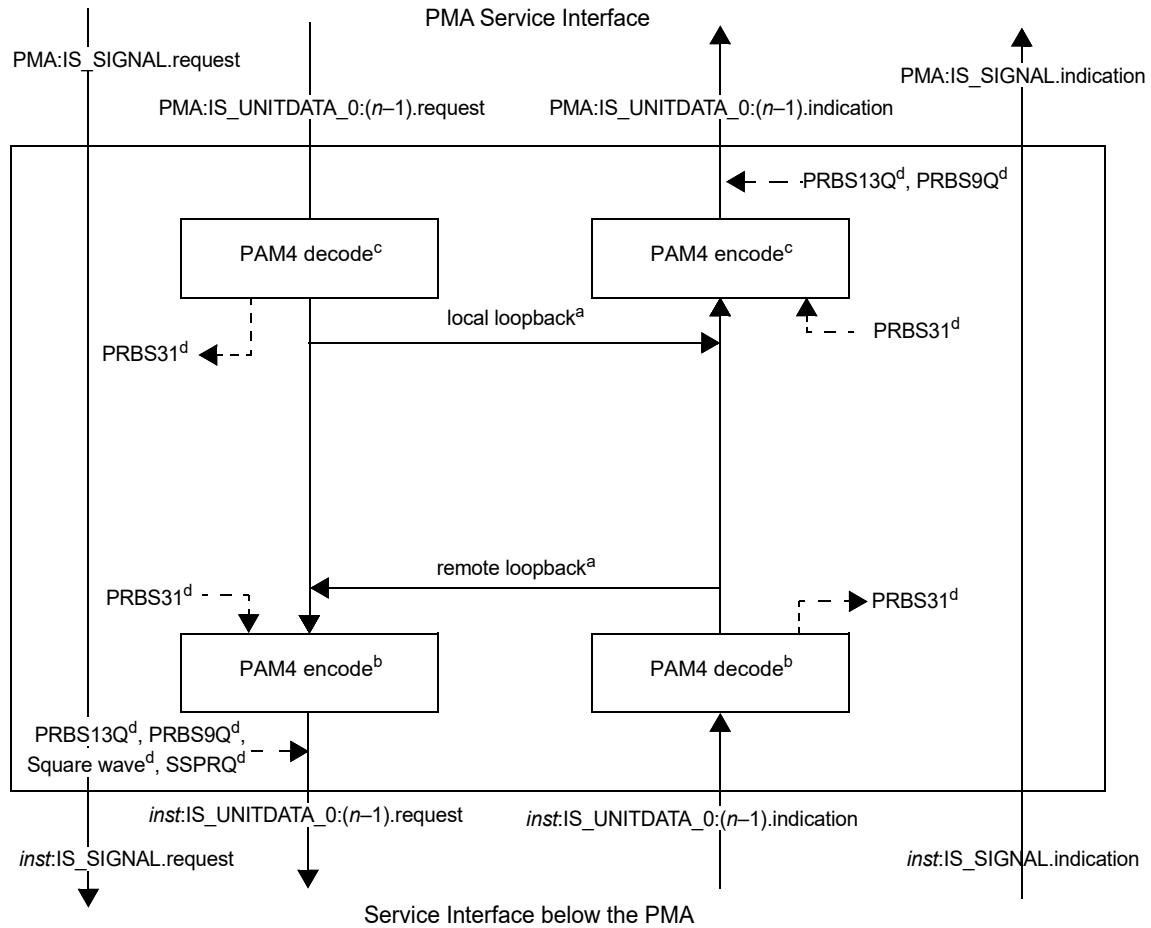


Figure 176–13—200GBASE-R 1:1, 400GBASE-R 2:2, 800GBASE-R 4:4, 1.6TBASE-R 8:8, 1.6TBASE-R 16:16 PMAs functional block diagram

176.7 Common functions within the PMA

176.7.1 PAM4 encoding

The PAM4 encoding function includes Gray mapping as specified in 176.7.1.1 and $1/(1+D) \bmod 4$ precoding as specified in 176.7.1.2.

176.7.1.1 Gray mapping

The Gray mapping for PAM4 encoded lanes is identical to that specified in 120.5.7.1.

176.7.1.2 Precoding

The precoding specifications in this subclause apply to the input and output lanes of a PMA that are connected to the service interface of an xBASE-KRn, xBASE-CRn, xBASE-DRn, or xBASE-FRn-500 PMD, or are part of an xAUI-n C2C or C2M link.

The PMA shall provide $1/(1+D) \bmod 4$ precoding capability on each output lane and may optionally provide $1/(1+D) \bmod 4$ decoding capability on each input lane. Precoding is implemented as specified in 135.5.7.2.

The precoder is enabled independently on the Tx output, Rx input, Rx output, and Tx input on each lane. Precoding is enabled and disabled using variables `precoder_tx_out_enablei`, `precoder_rx_in_enablei`, `precoder_rx_out_enablei`, and `precoder_tx_in_enablei`, where *i* is in the range 0 to (n-1).

If the PMA is connected to the service interface of an xBASE-KRn, xBASE-CRn, xBASE-DRn, or xBASE-FRn-500 PMD, or is part of an xAUI-n C2C or C2M link, and training is enabled by the management variable `mr_training_enable` (see 178B.12), then `precoder_tx_out_enablei` and `precoder_rx_in_enablei` shall be set as determined by the inter-sublayer link training (ILT) function in the `PATH_READY` state on lane *i* (see Figure 178B-10a and Figure 178B-10b). The method by which the ILT function affects these variables is implementation dependent.

If the PMA is connected to the service interface of an xBASE-KRn, xBASE-CRn, xBASE-DRn, or xBASE-FRn-500 PMD, or is part of an xAUI-n C2C or C2M link, and training is disabled by the management variable `mr_training_enable`, then `precoder_tx_out_enablei`, `precoder_rx_in_enablei`, `precoder_rx_out_enablei`, and `precoder_tx_in_enablei` are set as required by the implementation.

176.7.2 PMA local loopback mode (optional)

The PMA sublayer may provide a local loopback function. When local loopback mode is enabled, each bit received from the PMA:IS_UNITDATA_i.request(tx_symbol) primitive is looped back toward the direction of the MAC using the PMA:IS_UNITDATA_i.indication(rx_symbol) primitive.

During local loopback, the PMA continues to propagate data in the Tx direction.

The ability to perform this function is indicated by the `local_loopback_ability` PMA status variable. A device is placed in local loopback mode when the `local_loopback_enable` PMA control variable is set to one, and removed from local loopback mode when this variable is set to zero.

176.7.3 PMA remote loopback mode (optional)

Remote loopback, if provided, should be implemented in a PMA sublayer close enough to the PMD to maintain the bit sequence on each individual PMD lane. When remote loopback mode is enabled, each bit received from the *inst*:IS_UNITDATA_i.indication(rx_symbol) primitive is looped back toward the direction of the MDI using the *inst*:IS_UNITDATA_i.request(tx_symbol) primitive.

During remote loopback, the PMA continues to propagate data in the Rx direction.

The ability to perform this function is indicated by the `remote_loopback_ability` PMA status variable. A device is placed in remote loopback mode when the `remote_loopback_enable` PMA control variable is set to one, and removed from remote loopback mode when this variable is set to zero.

176.7.4 PMA test patterns

Where the lanes of the PMA are connected to a physically instantiated interface xAUI-n, Inner FEC, or a PMD service interface, the PMA generates and detects test patterns. These test patterns are used to test

adjacent layer interfaces for an individual PMA sublayer or to perform testing between a physically instantiated interface of a PMA and external testing equipment.

The test patterns, described below, are applied on individual output lanes of the PMA.

176.7.4.1 PRBS31 test pattern

The PMA shall include a PRBS31 pattern generator and checker as specified in 120.5.11.1.1. The PRBS31 pattern generator and checker is required when the PMA is adjacent to an xAUI-n, an Inner FEC, or a PMD.

176.7.4.2 PRBS31Q test pattern

The PRBS31Q test pattern generation is provided by the PRBS31 generator (see 176.7.4.1), followed by Gray mapping, and precoding (if enabled) in the PAM4 encoder (see 176.7.1).

The PRBS31Q test pattern checking is provided by the PRBS31 checker (see 176.7.4.1). PRBS31Q data is first processed by inverse precoding (if enabled) and inverse Gray mapped in the PAM4 decoder (see 176.4.3.5), prior to the PRBS31 pattern checker.

176.7.4.3 PRBS13Q test pattern

The PMA may include a PRBS13Q pattern generator as specified in 120.5.11.2.1, when the PMA is adjacent to a xAUI-n or a PMD.

176.7.4.4 PRBS9Q test pattern

The PMA may include a PRBS9Q pattern generator as specified in 120.5.11.2.a, when the PMA is adjacent to an xAUI-n or a PMD defined in Clause 178 or Clause 179.

176.7.4.5 SSPRQ test pattern

The PMA may include a SSPRQ pattern generator as specified in 120.5.11.2.3, when the PMA is adjacent to a PMD defined in Clause 180 or Clause 181.

176.7.4.6 Square wave (quaternary) test pattern

The PMA may include a square wave (quaternary) test pattern generator as specified in 120.5.11.2.4, when the PMA is adjacent to a PMD defined in Clause 180 or Clause 181.

176.7.4.7 PMA block error detection and counters

The PRBS31 test pattern checker in each PMAL (see 176.1.3) shall include block error detection and 17 related counters. Block error detection and behavior of the counters is defined in 174A.9.

The following counters are implemented per PMAL i , where $i = 0$ to $n-1$ and n is the number of PMALs:

test_block_error_bin_ i_k

A set of 16 48-bit counters where counter k counts once for each test block received with exactly k errored test symbols, $k = 0$ to 15.

test_block_error_bin_ i_{16p}

A 48-bit counter that counts once for each test block received with 16 or more errored test symbols.

176.8 Delay Constraints

There may be up to four instances of an SM-PMA within a Physical Layer, which is composed of an xBASE-R PHY and, optionally, an xMII Extender. The maximum delay (sum of transmit and receive delays) contributed by each instance of an xBASE-R PMA shall meet the values specified in Table 176–7. A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4, 169.4, and 174.4.

Table 176–7—Delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Maximum (ns)
200GBASE-R 1:8 PMA or 8:1 PMA	13 312	26	66.56
200GBASE-R 1:1 PMA	3 072	6	15.36
400GBASE-R 2:16 PMA or 16:2 PMA	16 384	32	40.96
400GBASE-R 2:2 PMA	6 144	12	15.36
800GBASE-R 4:32 PMA or 32:4 PMA	12 288	24	15.36
800GBASE-R 4:4 PMA	12 288	24	15.36
1.6TBASE-R 8:16 PMA or 16:8 PMA	24 576	48	15.36
1.6TBASE-R 8:8 PMA or 16:16 PMA	24 576	48	15.36

176.9 Skew Constraints

The Skew (relative delay) between the PCSs is kept within limits so that the information on the lanes can be reassembled by the PCS.

Any PMA that combines PCSs from different input lanes onto the same output lane is required to tolerate Skew Variation between the input lanes without changing the PCSL positions on the output lane.

176.9.1 Skew constraints for 200GBASE-R and 400GBASE-R SM-PMAs

Skew and Skew Variation are defined in 116.5 and specified at the points SP1 to SP6 shown in Figure 116–5.

If the PMA service interface is physically instantiated so that the Skew at SP1 and SP6 can be measured, then the Skew at SP1 is limited to 29 ns, the Skew Variation at SP1 is limited to 200 ps, the Skew at SP6 is limited to 160 ns and the Skew Variation at SP6 is limited to 3.8 ns.

176.9.2 Skew constraints for 800GBASE-R SM-PMAs

Skew and Skew Variation are defined in 169.5 and specified at the points SP1 to SP6 as shown in Figure 169–4 and Figure 169–5.

If the PMA service interface is physically instantiated so that the Skew at SP1 and SP6 can be measured, then the Skew at SP1 is limited to 16 ns, the Skew Variation at SP1 is limited to 200 ps, the Skew at SP6 is limited to 145 ns and the Skew Variation at SP6 is limited to 3.8 ns.

176.9.3 Skew constraints for 1.6TBASE-R SM-PMAs

Skew and Skew Variation are defined in 174.5 and specified at the points SP1 to SP6 as shown in Figure 174–5 and Figure 174–6.

If the PMA service interface is physically instantiated so that the Skew at SP1 and SP6 can be measured, then the Skew at SP1 is limited to 16 ns, the Skew Variation at SP1 is limited to 200 ps, the Skew at SP6 is limited to 145 ns and the Skew Variation at SP6 is limited to 3.8 ns.

176.10 Path data delay for time synchronization

The SM-PMA sublayer optionally provides path data delay information for time synchronization (see Clause 90).

The ability to report SM-PMA path data delays is indicated by the `PMA_delay_ns_TX_ability`, `PMA_delay_subns_TX_ability`, `PMA_delay_ns_RX_ability`, and `PMA_delay_subns_RX_ability` status variables (see Table 176–9).

The SM-PMA sublayers have an intrinsic delay where the sum of the transmit and receive delays is a constant value. The transmit path data delay for the m:n and n:n PMAs are reported as the delay incurred by the first bit of the PCS alignment marker of the PCS lane with the largest delay through the PMA. The transmit path data delay for the n:m PMA is reported as the delay incurred by the first bit of the PCS alignment marker of the PCS lane with the smallest delay through the PMA. The receive path data delay for the m:n and n:n PMAs are reported as the delay incurred by the first bit of the PCS alignment marker of the PCS lane with the smallest delay through the PMA. The receive path data delay for the n:m PMA is reported as the delay incurred by the first bit of the PCS alignment marker of the PCS lane with the largest delay through the PMA. To determine the PCS lanes with the largest and smallest delay, delays added to PCS lanes (see 176.4.2.4 and 176.4.3.3) as well as symbol multiplexing and demultiplexing delays (see 176.4.2.5 and 176.4.3.2) shall be considered.

Four separate delays are reported in the following eight path data delay status variables:

- `PMA_delay_ns_TX_max` and `PMA_delay_subns_TX_max`
- `PMA_delay_ns_TX_min` and `PMA_delay_subns_TX_min`
- `PMA_delay_ns_RX_max` and `PMA_delay_subns_RX_max`
- `PMA_delay_ns_RX_min` and `PMA_delay_subns_RX_min`

176.11 Management variables

PMA control and status variables intended to be accessible via a management system are listed in Table 176–8 and Table 176–9.

If the MDIO interface (see Clause 45) is implemented, PMA control and status variables shall be mapped to MDIO registers as shown in Table 176–8 and Table 176–9. These tables provide the MDIO register and bit numbers for the PMA addressed as MMD 1. For implementations with multiple PMA sublayers, additional PMA sublayers use the corresponding register and bit numbers in MMDs 8, 9, 10, and 11 as necessary

If the MDIO interface is not implemented, an alternate mechanism to access management variables shall be provided.

Table 176–8—PMA control variables and MDIO mapping

Control variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
Remote_loopback_enable	176.7.3	1.0.1	45.2.1.1.4
Local_loopback_enable	176.7.2	1.0.0	45.2.1.1.5
precoder_tx_out_enable_<0:15>	176.7.1.2	1.600.0 to 1.600.15	45.2.1.139
precoder_rx_in_enable_<0:15>	176.7.1.2	1.601.0 to 1.601.15	45.2.1.140
precoder_rx_out_enable_<0:15>	176.7.1.2	1.602.0 to 1.602.15	45.2.1.141
precoder_tx_in_enable_<0:15>	176.7.1.2	1.603.0 to 1.603.15	45.2.1.142
PRBS31_pattern_enable	176.7.4.1	1.1501.7	45.2.1.170
SSPRQ_pattern_enable	176.7.4.5	1.1501.14	45.2.1.170
PRBS_Tx_gen_enable	176.7.4.2, 176.7.4.3	1.1501.3	45.2.1.170
PRBS_Tx_check_enable	176.7.4.2, 176.7.4.3	1.1501.2	45.2.1.170
PRBS_Rx_gen_enable	176.7.4.2, 176.7.4.3	1.1501.1	45.2.1.170
PRBS_Rx_check_enable	176.7.4.2, 176.7.4.3	1.1501.0	45.2.1.170
Square_wave_enable_<0:15>	176.7.4.6	1.1510.0 to 1.1510.15	45.2.1.171
PRBS9Q_enable_<0:7>	176.7.4.4	1.1511.0 to 1.1511.7	45.2.1.171a
PRBS13Q_enable_<0:7>	176.7.4.3	1.1512.0 to 1.1512.7	45.2.1.172
PMA_reset	176.4.4.2.1	1.0.15	45.2.1.1

Table 176–9—PMA status variables and MDIO mapping

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
800G_Remote_loopback_ability	176.7.3	1.73.15	45.2.1.60b.100
Local_loopback_ability	176.7.2	1.8.0	45.2.1.7.15
precoder_rx_in_request_flag	176.7.1.2	1.604.0	45.2.1.143.2
precoder_tx_in_request_flag	176.7.1.2	1.604.1	45.2.1.143.1
precoder_rx_in_request_<0:7>	176.7.1.2	1.605.0 to 1.605.7	45.2.1.144
precoder_tx_in_request_<0:7>	176.7.1.2	1.606.0 to 1.606.7	45.2.1.145
Square_wave_ability	176.7.4.6	1.1500.12	45.2.1.169

Table 176–9—PMA status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
PRBS13Q_gen_Tx_ability	176.7.4.3	1.1500.11	45.2.1.169
PRBS13Q_gen_Rx_ability	176.7.4.3	1.1500.10	45.2.1.169
SSPRQ_gen_Tx_ability	176.7.4.5	1.1500.13	45.2.1.169
PRBS9Q_gen_Rx_ability	176.7.4.4	1.1500.14	45.2.1.169
PRBS9Q_gen_Tx_ability	176.7.4.4	1.1500.15	45.2.1.169
Ln<0:7>_PRBS_Tx_test_err_counter	176.7.4.2	1.1600 to 1.1607	45.2.1.173
Ln<0:7>_PRBS_Rx_test_err_counter	176.7.4.2	1.1700 to 1.1707	45.2.1.174
test_block_error_bin_tx_0<0:15>	176.7.4.7	1.2900 to 1.2947	45.2.1.272
test_block_error_bin_tx_0_16p	176.7.4.7	1.2948 to 1.2950	45.2.1.272
test_block_error_bin_tx_1<0:15>	176.7.4.7	1.2951 to 1.2998	45.2.1.272
test_block_error_bin_tx_1_16p	176.7.4.7	1.2999 to 1.3001	45.2.1.272
test_block_error_bin_tx_2<0:15>	176.7.4.7	1.3002 to 1.3049	45.2.1.272
test_block_error_bin_tx_2_16p	176.7.4.7	1.3050 to 1.3052	45.2.1.272
test_block_error_bin_tx_3<0:15>	176.7.4.7	1.3053 to 1.3100	45.2.1.272
test_block_error_bin_tx_3_16p	176.7.4.7	1.3101 to 1.3103	45.2.1.272
test_block_error_bin_tx_4<0:15>	176.7.4.7	1.3104 to 1.3151	45.2.1.272
test_block_error_bin_tx_4_16p	176.7.4.7	1.3152 to 1.3154	45.2.1.272
test_block_error_bin_tx_5<0:15>	176.7.4.7	1.3155 to 1.3202	45.2.1.272
test_block_error_bin_tx_5_16p	176.7.4.7	1.3203 to 1.3205	45.2.1.272
test_block_error_bin_tx_6<0:15>	176.7.4.7	1.3206 to 1.3253	45.2.1.272
test_block_error_bin_tx_6_16p	176.7.4.7	1.3254 to 1.3256	45.2.1.272
test_block_error_bin_tx_7<0:15>	176.7.4.7	1.3257 to 1.3304	45.2.1.272
test_block_error_bin_tx_7_16p	176.7.4.7	1.3305 to 1.3307	45.2.1.272
test_block_error_bin_rx_0<0:15>	176.7.4.7	1.3308 to 1.3355	45.2.1.272
test_block_error_bin_rx_0_16p	176.7.4.7	1.3356 to 1.3358	45.2.1.272
test_block_error_bin_rx_1<0:15>	176.7.4.7	1.3359 to 1.3406	45.2.1.272
test_block_error_bin_rx_1_16p	176.7.4.7	1.3407 to 1.3409	45.2.1.272
test_block_error_bin_rx_2<0:15>	176.7.4.7	1.3410 to 1.3457	45.2.1.272
test_block_error_bin_rx_2_16p	176.7.4.7	1.3458 to 1.3460	45.2.1.272
test_block_error_bin_rx_3<0:15>	176.7.4.7	1.3461 to 1.3508	45.2.1.272
test_block_error_bin_rx_3_16p	176.7.4.7	1.3509 to 1.3511	45.2.1.272
test_block_error_bin_rx_4<0:15>	176.7.4.7	1.3512 to 1.3559	45.2.1.272
test_block_error_bin_rx_4_16p	176.7.4.7	1.3560 to 1.3562	45.2.1.272
test_block_error_bin_rx_5<0:15>	176.7.4.7	1.3563 to 1.3610	45.2.1.272

Table 176–9—PMA status variables and MDIO mapping (*continued*)

Status variable	Variable reference	MDIO register/bit number	MDIO register/bit reference
test_block_error_bin_rx_5_16p	176.7.4.7	1.3611 to 1.3613	45.2.1.272
test_block_error_bin_rx_6_<0:15>	176.7.4.7	1.3614 to 1.3661	45.2.1.272
test_block_error_bin_rx_6_16p	176.7.4.7	1.3662 to 1.3664	45.2.1.272
test_block_error_bin_rx_7_<0:15>	176.7.4.7	1.3665 to 1.3712	45.2.1.272
test_block_error_bin_rx_7_16p	176.7.4.7	1.3713 to 1.3715	45.2.1.272
align_status_mux	176.4.4.2.1	1.1480.13	45.2.1.169g
all_locked_mux	176.4.4.2.1	1.1480.12	45.2.1.169g
deskew_done_mux	176.4.4.2.1	1.1480.11	45.2.1.169g
pcs_lanes_identified_mux	176.4.4.2.1	1.1480.10	45.2.1.169g
all_locked_demux	176.4.4.2.1	1.1480.9	45.2.1.169g
pcs_lanes_identified_demux	176.4.4.2.1	1.1480.8	45.2.1.169g
pmal_locked_demux<7:0>	176.4.4.2.1	1.1480.7:0	45.2.1.169g
PMA_delay_ns_TX_max	176.10	1.1801, 1.1802	45.2.1.176
PMA_delay_subns_TX_max	176.10	1.1809	45.2.1.176
PMA_delay_ns_TX_min	176.10	1.1803, 1.1804	45.2.1.176
PMA_delay_subns_TX_min	176.10	1.1810	45.2.1.176
PMA_delay_ns_RX_max	176.10	1.1805, 1.1806	45.2.1.177
PMA_delay_subns_RX_max	176.10	1.1811	45.2.1.177
PMA_delay_ns_RX_min	176.10	1.1807, 1.1808	45.2.1.177
PMA_delay_subns_RX_min	176.10	1.1812	45.2.1.177
PMA_delay_ns_RX_ability	176.10	1.1800.0	45.2.1.175
PMA_delay_ns_TX_ability	176.10	1.1800.1	45.2.1.175
PMA_delay_subns_RX_ability	176.10	1.1800.2	45.2.1.175
PMA_delay_subns_TX_ability	176.10	1.1800.3	45.2.1.175

176.12 Protocol implementation conformance statement (PICS) proforma for Clause 176, Physical Medium Attachment (PMA) sublayer with RS-FEC symbol multiplexing, type 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R⁶

176.12.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 176, Physical Medium Attachment (PMA) sublayer with RS-FEC symbol multiplexing, type 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

176.12.2 Identification

176.12.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

176.12.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3dj-202x, Clause 176, Physical Medium Attachment (PMA) sublayer with RS-FEC symbol multiplexing, type 200GBASE-R, 400GBASE-R, 800GBASE-R, and 1.6TBASE-R
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3dj-202x.)	

Date of Statement	
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⁶*Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

176.12.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
P1	200GBASE-R 8:1 PMA	176.4		O/1	Yes [] No []
P2	400GBASE-R 16:2 PMA	176.4		O/1	Yes [] No []
P3	800GBASE-R 32:4 PMA	176.4		O/1	Yes [] No []
P4	1.6TBASE-R 16:8 PMA	176.4		O/1	Yes [] No []
P5	200GBASE-R 1:8 PMA	176.5		O/1	Yes [] No []
P6	400GBASE-R 2:16 PMA	176.5		O/1	Yes [] No []
P7	800GBASE-R 4:32 PMA	176.5		O/1	Yes [] No []
P8	1.6TBASE-R 8:16 PMA	176.5		O/1	Yes [] No []
P9	200GBASE-R 1:1 PMA	176.6		O/1	Yes [] No []
P10	400GBASE-R 2:2 PMA	176.6		O/1	Yes [] No []
P11	800GBASE-R 4:4 PMA	176.6		O/1	Yes [] No []
P12	1.6TBASE-R 8:8 PMA	176.6		O/1	Yes [] No []
P13	1.6TBASE-R 16:16 PMA	176.6		O/1	Yes [] No []
*AUI	AUI directly above or below the PMA	176.1.5		O.2	Yes [] No []
*IFEC	Inner FEC directly below the PMA	176.1.5		O.2	Yes [] No []
*PMD	PMD directly below the PMA	176.1.5		O.2	Yes [] No []
*PMDO	Optical PMD directly below the PMA	176.1.5		PMD:O	Yes [] No []
*PMDE	Electrical PMD directly below the PMA	176.1.5		PMD:O	Yes [] No []
SIG1	SIGNAL_OK generation above the PMA	176.2		M	Yes []
SIG2	SIGNAL_OK generation below the PMA	176.3		M	Yes []
PE1	PAM4 encoding (Gray mapping and precoding)	176.7.1		AUI + PMD:M	Yes [] N/A []
*MD	MDIO	176.11		O	Yes [] No []

176.12.3.1 Loopback

Item	Feature	Subclause	Value/Comment	Status	Support
L1	PMA local loopback	176.7.2		O	Yes [] No []
L2	PMA remote loopback	176.7.3		O	Yes [] No []

176.12.3.2 Test patterns

Item	Feature	Subclause	Value/Comment	Status	Support
T1	PRBS31	176.7.4.1		AUI + PMD+ IFEC:M	Yes [] N/A []
T2	PRBS31Q	176.7.4.2		AUI + PMD:M	Yes [] N/A []
T3	PRBS13Q	176.7.4.3		AUI + PMD:O	Yes [] No [] N/A []
T4	PRBS9Q	176.7.4.4		AUI + PMDE: O	Yes [] No [] N/A []
T5	SSPRQ	176.7.4.5		PMDO* !IFEC:O	Yes [] No [] N/A []
T6	Square wave	176.7.4.6		PMDO* !IFEC:O	Yes [] No [] N/A []

176.12.3.3 Delay constraints

Item	Feature	Subclause	Value/Comment	Status	Support
D1	PMA Delay constraints	176.8		M	Yes []

176.12.3.4 Skew constraints

Item	Feature	Subclause	Value/Comment	Status	Support
S1	PMA Skew constraints	176.9		M	Yes []

176.12.3.5 Time synchronization

Item	Feature	Subclause	Value/Comment	Status	Support
*TS	Support for time synchronization	176.10		O	Yes [] No []
TS1	Path data delays	176.10	Path data delay reported for the transmit path and for the receive path through the PMA.	TS:M	Yes [] N/A []
TS2	Nanosecond resolution	176.10	Path data delays reported with nanosecond or better resolution.	TS:M	Yes [] N/A []
TS3	Sub-nanosecond resolution	176.10	Path data delays reported with sub-nanosecond resolution.	TS:O	Yes [] No [] N/A []

176.12.3.6 Management

Item	Feature	Subclause	Value/Comment	Status	Support
M1	Alternate access to management variables is provided	176.11		!MD:M	Yes [] N/A []
M2	Mapping of MDIO control bits and MDIO status bits	176.11		MD:M	Yes [] N/A []